Predicting E-Cigarette Use Among Thai Youth Using Machine Learning: The Role of Mental Health and Risk Behaviors

Thanathon Yichaturat, Petsinee Trongpetch, Waraporn Kaewkui, Warisara Mongkholwat and Sirinya Teeraananchai
Department of Statistics, Faculty of Science, Kasetsart University, Bangkok 10900, Thailand

Abstract

The use of e-cigarettes among Thai youth is increasing and becoming a public health concern. This study aimed to evaluate five machine learning models to predict e-cigarette use behavior among Thai youth. The data were balanced using three techniques, and the ADASYN method provided the best results. The Random Forest model showed the highest performance, with a sensitivity of 78.95% and an AUC of 72.90%. Key factors included Coping Strategies for stress (With alcohol). These findings suggest that Random Forest can help identify risky behaviors and support prevention efforts.

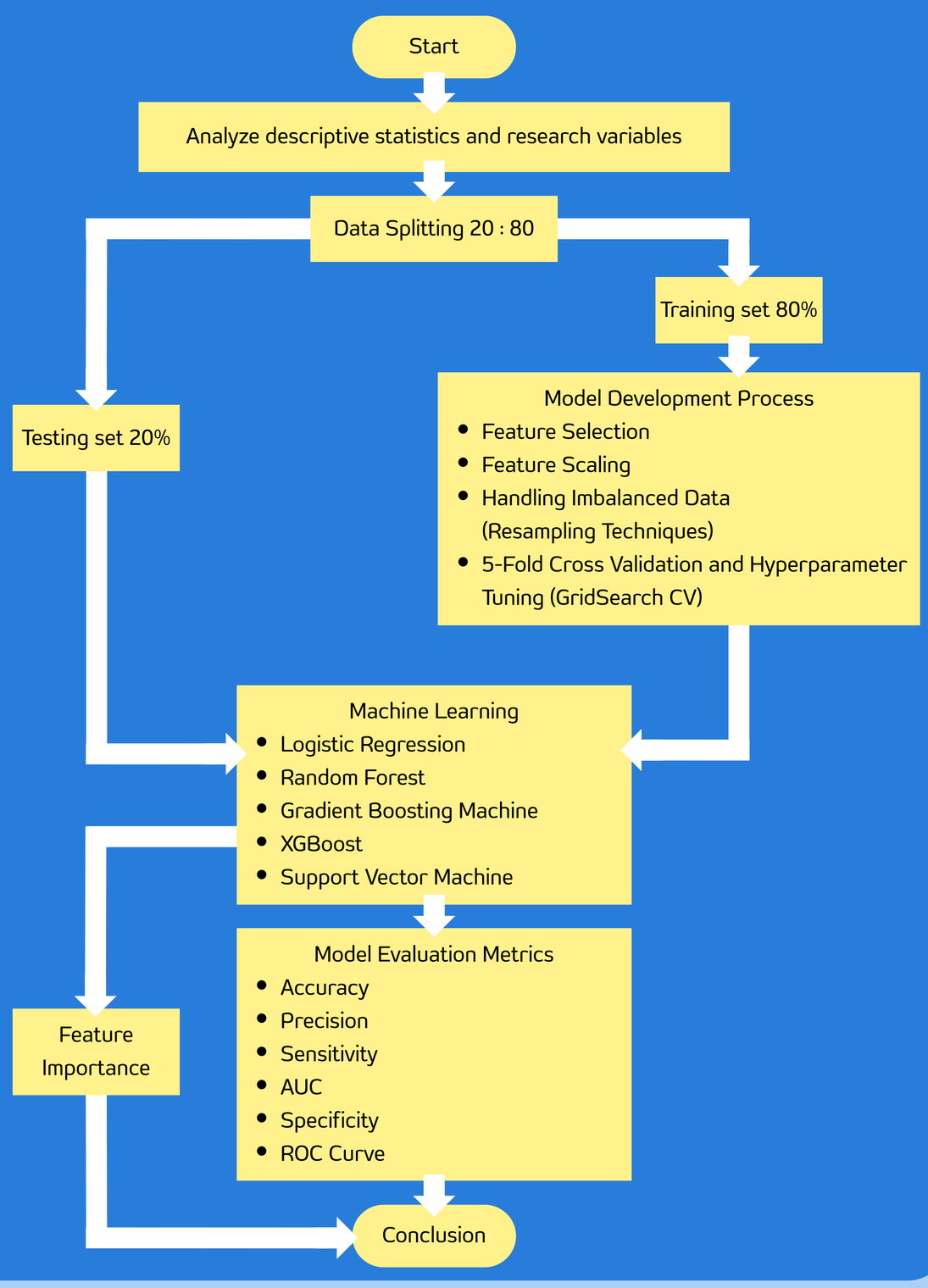
Introduction

Currently, the usage of e-cigarettes among Thai youth is on a steady and alarming growing trend. According to data from Thailand's National Statistical Office (2022), the number of e-cigarette users has increased by over 10 percent in just one year. The situation demonstrates that e-cigarette use among Thai youths and teens is concerning and rapidly increasing, necessitating close monitoring and legislative action to minimize long-term health implications.

Objective

To evaluate the performance across five machine learning models in predicting e-cigarette use behavior among Thai youth, including Logistic Regression, Support Vector Machines, Random Forest, Gradient Boosting Machines and XGBoost to examine the key factors influencing e-cigarette use behavior among Thai youth.

Materials and Methods



Acknowledgments

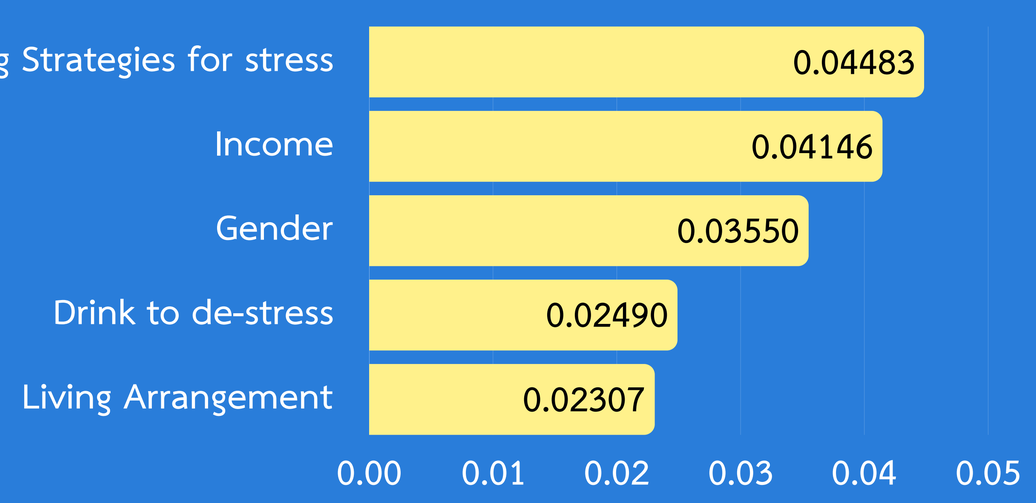
This project was supported by the Department of Statistics, Kasetsart University and the Undergraduate Research Matching Fund, Faculty of Science, Kasetsart University. Special thanks are extended to Assoc. Prof. Dr. Sirinya Teeraananchai, the project advisor, for her expert guidance and insightful recommendations, which contributed to the successful completion of this project.

Result and Discussion

Table 1 Comparison table of model performance using Original data and data processed with Adaptive Synthetic Sampling (ADASYN) and Cross-Validation (CV).

Model	Original (%)			With ADASYN and CV (%)		
	Sensitivity	Specificity	AUC	Sensitivity	Specificity	AUC
LR	78.95	62.81	72.59	57.89	75.21	69.99
RF	73.68	60.33	67.86	78.95	67.77	72.90
GBM	31.58	95.87	67.81	63.61	71.07	67.64
XGBoost	52.63	83.47	65.00	57.89	78.51	68.51
SVM	78.94	59.50	71.34	31.58	95.04	70.68

Fig1. SHAP impact bar plot of feature contributions for the Random Forest model using ADASYN and cross-validation.



The analysis of e-cigarette use behavior among Thai youth using machine learning models. The results showed that the Random Forest model provided the best predictive performance when combined with LASSO feature selection, ADASYN data balancing and 5-fold stratified cross-validation, leading to an improvement in AUC. The most important factors related to e-cigarette use were coping strategies for dealing with stress, especially among alcohol drinkers, and respondents' income.

Conclusion

Overall Best Performance: After applying ADASYN, the model improved its ability to detect e-cigarette users, with Sensitivity increasing from 73.68% to 78.95% and AUC from 67.86% to 72.90%. In addition, Specificity also improved from 60.33% to 67.77%, indicating better overall performance compared to the original imbalanced data.

Reference

Senate. 2025. Proactive measures to prevent and address the problem of e-cigarettes among children and youth. Bangkok, Thailand. https://senate.go.th/view/386/News/SenateMagazine/283/TH-TH. 2 August 2025

Supaknapha J., Natcha B., Benyathip K., and Nattapat P. 2024. Impact of Mental Health on Alcohol Consumption Behavior among Adolescents in Bangkok. Bangkok, Thailand, pp.30-31

Fang, W., Liu, Y., Xu, C., Luo, X. & Wang, K. 2024. Feature Selection and Machine Learning Approaches in Prediction of Current E-Cigarette Use Among U.S. Adults in 2022. Int. J. Environ. Res. Public Health 2024, 21(11), 1474; doi.org/10.3390/ijerph 21111474